

Dynamic Multi-Tenant Form Management & Reporting Platform with Conditional Workflow Engine

LLD

1. System Modules

Auth Module

- |
- |—— Registration
- |—— Login
- |—— OTP Verification
- |—— JWT Authentication

User Management Module

- |
- |—— Super Admin
- |—— Center
- |—— Organization
- |—— User

Form Management Module

- |
- |—— Form Builder
- |—— Form Renderer
- |—— Form Distribution
- |—— Form Versioning
- |—— Form Cloning

Conditional Logic Module

- |
- |—— Question Tree
- |—— Rule Engine
- |—— Dynamic Rendering

Submission Module

- |
- |—— Save Draft
- |—— Submit Form
- |—— Data Validation

Report Module

- |
- |—— PDF Report
- |—— Excel Report

├── HTML Report
└── Dynamic Filters

Notification Module

├── Email OTP
├── Email Notifications
└── System Alerts

2. Database LLD

User Collection

```
{  
  _id:ObjectId,  
  
  firstName:String,  
  lastName:String,  
  
  email:String,  
  phone:String,  
  
  password:String,  
  
  role:{  
    type:String,  
    enum:[  
      "SUPER_ADMIN",  
      "CENTER",  
      "ORGANIZATION",  
      "USER"  
    ]  
  },  
  
  centerId:ObjectId,  
  
  organizationId:ObjectId,  
  
  isVerified:Boolean,  
  
  status:{  
    type:String,  
    enum:[  
      "ACTIVE",  
      "INACTIVE",  
      "BLOCKED"  
    ]  
  },  
  
  createdAt:Date,
```

```
    updatedAt:Date
}
```

Center Collection

```
{
  _id:ObjectId,

  name:String,

  description:String,

  status:String,

  createdBy:ObjectId,

  createdAt:Date
}
```

Organization Collection

```
{
  _id:ObjectId,

  centerId:ObjectId,

  name:String,

  description:String,

  status:String,

  restrictedForms:[ObjectId],

  createdAt:Date
}
```

3. Form Builder LLD

Form Collection

```
{
  _id:ObjectId,

  title:String,

  description:String,

  visibility:String,

  version:Number,
```

```
status:String,  
createdBy:ObjectId,  
publishedAt:Date,  
createdAt:Date  
}
```

Section Collection

```
{  
  _id:ObjectId,  
  formId:ObjectId,  
  title:String,  
  order:Number  
}
```

Question Collection

```
{  
  _id:ObjectId,  
  formId:ObjectId,  
  sectionId:ObjectId,  
  label:String,  
  type:String,  
  required:Boolean,  
  order:Number,  
  validations:[]  
}
```

Field Types

```
TEXT  
TEXTAREA  
NUMBER  
EMAIL  
PHONE  
DATE  
TIME
```

RADIO
CHECKBOX
DROPDOWN
FILE
SIGNATURE

4. Validation LLD

Validation Object

```
{  
  minLength:3,  
  
  maxLength:50,  
  
  minValue:1,  
  
  maxValue:100,  
  
  regex:"",  
  
  required:true  
}
```

Example:

```
{  
  field:"Age",  
  
  validation:{  
    minValue:18,  
    maxValue:60  
  }  
}
```

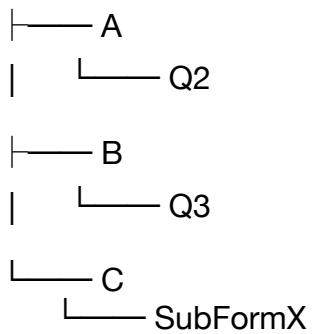
5. Conditional Question Engine

Question Tree Schema

```
{  
  _id:ObjectId,  
  
  parentQuestionId:ObjectId,  
  
  triggerOptionId:ObjectId,  
  
  childQuestionId:ObjectId  
}
```

Example:

Q1



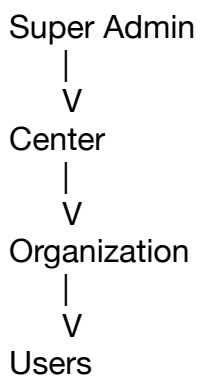
Renderer Logic

```
if(answer === "A")  
{  
  render(Q2)  
}
```

```
if(answer === "B")  
{  
  render(Q3)  
}
```

6. Form Distribution LLD

Public Form



Form Assignment Collection

```
{  
  _id:ObjectId,  
  
  formId:ObjectId,  
  
  centerId:ObjectId,  
  
  organizationId:ObjectId,
```

```
assignedBy:ObjectId,  
assignedAt:Date  
}
```

7. Organization Form Clone

Never edit original form.

Create a copy.

Master Form

|
V

Clone

|
V

Organization Form

Form Clone Schema

```
{  
  _id:ObjectId,  
  masterFormId:ObjectId,  
  organizationId:ObjectId,  
  version:Number  
}
```

8. Submission Module LLD

Submission Collection

```
{  
  _id:ObjectId,  
  formId:ObjectId,  
  userId:ObjectId,  
  organizationId:ObjectId,  
  status:String,  
  submittedAt:Date,  
  answers:[]  
}
```

Answer Structure

```
{  
  questionId:ObjectId,  
  
  answer:"Yes"  
}
```

9. OTP Module LLD

OTP Collection

```
{  
  _id:ObjectId,  
  
  email:String,  
  
  otp:String,  
  
  expiresAt:Date,  
  
  verified:Boolean  
}
```

OTP Flow

```
Register  
|  
Generate OTP  
|  
Save OTP  
|  
Email OTP  
|  
Verify OTP  
|  
Activate Account
```

10. Report Module LLD

Report Collection

```
{  
  _id:ObjectId,  
  
  formId:ObjectId,  
  
  generatedBy:ObjectId,
```



```
reportType:String,  
  
filters:{},  
  
generatedAt:Date  
}
```

Report Types

HTML

PDF

EXCEL

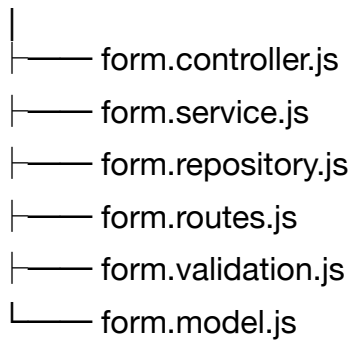
CSV

11. Backend LLD

```
src  
|  
|—— modules  
|  
|  |—— auth  
|  |  
|  |—— users  
|  |  
|  |—— centers  
|  |  
|  |—— organizations  
|  |  
|  |—— forms  
|  |  
|  |—— submissions  
|  |  
|  |—— reports  
|  |  
|  |—— notifications  
|—— shared  
|  
|—— middleware  
|  
|—— config  
|  
|—— app.js
```

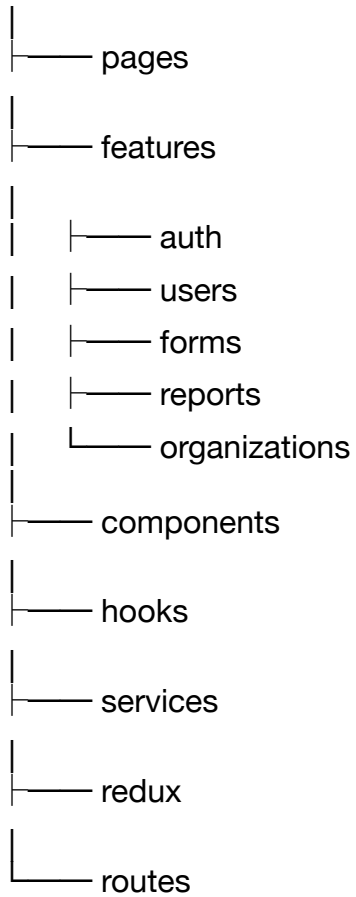
Module Structure

forms

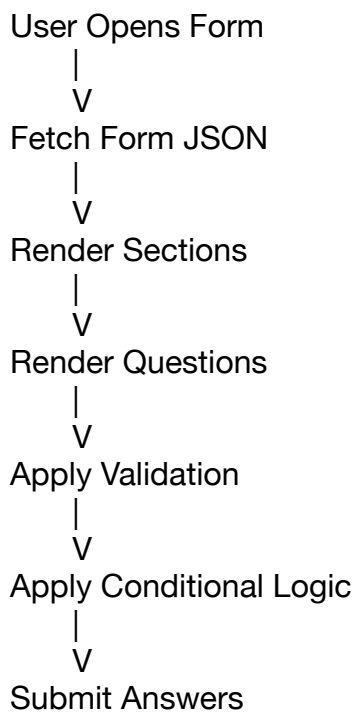


12. Frontend LLD

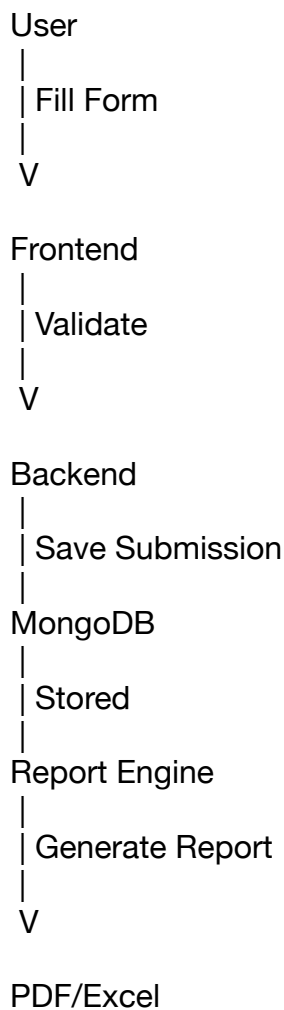
src



13. Dynamic Form Rendering Flow



14. Sequence Diagram (Form Submission)

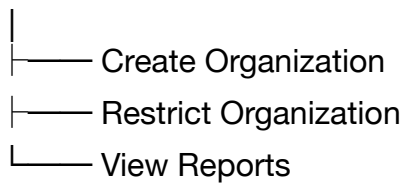


15. RBAC (Role-Based Access Control)

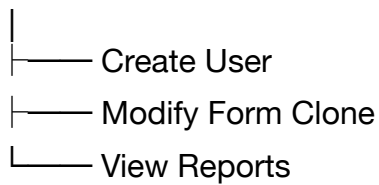
SUPER_ADMIN



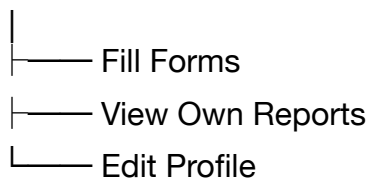
CENTER



ORGANIZATION



USER



Suggested Design Pattern

Use:

Controller Layer



Service Layer



Repository Layer



MongoDB

This keeps the code modular, testable, scalable, and easy to maintain.